

Independence of Events: Let $(\Omega, \mathcal{F}, P)$ be a probability space, and let $A, B \in \mathcal{F}$, with $P(B) > 0$. By the multiplication rule we have

$$P(A \cap B) = P(B)P(A|B).$$

In many experiments the information provided by B does not affect the probability of event A , i.e., $P(A|B) = P(A)$. Eg., winning the toss by the Indian cricket team captain in consecutive matches are independent events. Whereas, let event A be the life of an equipment exceeding 100 hours and event B be life of the equipment exceeding 200 hours. Then events A and B are dependent events.

Two events, A and B are said to be independent if and only if

$$P(A \cap B) = P(A)P(B).$$

Remark 1: We have not placed any restriction on $P(A)$ or $P(B)$. Thus conditional probability is not defined when $P(A)$ or $P(B) = 0$, but independence is. Clearly, if $P(A) = 0$, then A is independent of every $E \in \mathcal{F}$. Also, any event $A \in \mathcal{F}$ is independent of ϕ and Ω .

Theorem 1: If A and B are independent events, then

$$P(A|B) = P(A), \text{ if } P(B) > 0,$$

$$P(B|A) = P(B), \text{ if } P(A) > 0.$$

Theorem 2: If A and B are independent events, so are A and B^c , A^c and B , and A^c and B^c .

Proof: The proof of all three cases is provided below:

$$\begin{aligned} P(A \cap B^c) &= P(A - (A \cap B)) \\ &= P(A) - P(A \cap B) \text{ [since } (A \cap B) \subseteq A] \\ &= P(A)[1 - P(B)] \\ &= P(A)P(B^c). \end{aligned}$$

$$\begin{aligned}
P(A^c \cap B) &= P(B - (A \cap B)) \\
&= P(B) - P(A \cap B) \quad [\text{since } (A \cap B) \subseteq B] \\
&= P(B)[1 - P(A)] \\
&= P(A^c)P(B).
\end{aligned}$$

$$\begin{aligned}
P(A^c \cap B^c) &= 1 - P(A \cup B) \quad [\text{De Morgan's Law}] \\
&= 1 - [P(A) + P(B) - P(A \cap B)] \quad [\text{Law of Addition}] \\
&= 1 - P(A) - P(B) + P(A)P(B) \quad [A \text{ and } B \text{ are independent}] \\
&= [1 - P(A)] - P(B)[1 - P(A)] \\
&= [1 - P(A)][1 - P(B)] \\
&= P(A^c)P(B^c).
\end{aligned}$$

Remark 2: If two events (each with nonzero probability) are mutually exclusive, they are not independent since the occurrence of one will automatically preclude the occurrence of the other. Similarly if two events are independent (each with nonzero probability), then they can't be mutually exclusive.

Remark 2: Three events A, B, C are independent, if

$$\begin{aligned}
P(A \cap B) &= P(A)P(B), \\
P(A \cap C) &= P(A)P(C), \\
P(B \cap C) &= P(B)P(C), \\
P(A \cap B \cap C) &= P(A)P(B)P(C).
\end{aligned}$$

Remark 3: “ n ” Events $A_1, A_2, \dots, A_n$ are independent, if

$$\begin{aligned}
P(A_i \cap A_j) &= P(A_i)P(A_j), \quad i \neq j; \quad i, j = 1, 2, \dots, n, \\
P(A_i \cap A_j \cap A_k) &= P(A_i)P(A_j)P(A_k), \quad i \neq j \neq k; \quad i, j, k = 1, 2, \dots, n, \\
&\vdots \\
P(A_1 \cap A_2 \cap \dots \cap A_n) &= P(A_1)P(A_2)\dots P(A_n).
\end{aligned}$$

The total number of conditions to determine the independence of “ n ” events is $2^n - n - 1$.

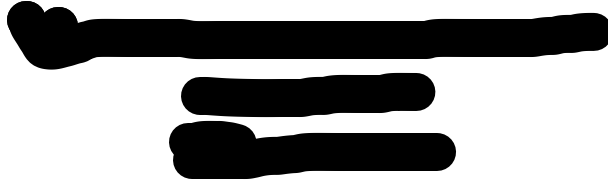
Remark 4: Independence implies pairwise independence, but the converse is not true.

Example 1: Two machines, A and B operate independently. The probability that machine A fails on a given day is 0.01, and the probability that machine B fails on a given day is 0.02. Let E_1 be the event that machine A fails on a given day, and E_2 be the event that machine B fails on the same day. Determine if events E_1 and E_2 are independent.

Solution: We have $P(E_1) = 0.01$ and $P(E_2) = 0.02$, and $P(E_1 \cap E_2) = P(E_1)P(E_2) = 0.01 \times 0.02 = 0.0002$. Hence E_1 and E_2 are independent.

Example 2: In a “Probability and Statistics” course exam, Student 'A' can solve 75% of the problems and Student 'B' can solve 70% of the problems. Both the students solve the problems independently. What is the probability that either A or B can solve a randomly selected problem?

Solution: $P(A) = 0.75, P(B) = 0.70, P(A \cap B) = P(A)P(B) = 0.75 \times 0.70 = 0.525$. We have to find $P(A \cup B)$, which is given by $P(A \cup B) = P(A) + P(B) - P(A \cap B) = P(A) + P(B) - P(A)P(B) = 0.75 + 0.70 - 0.525 = 0.925$.



Random Variable: Have a look at the following scenarios:

1. A bank may like to know the number of fraudulent transactions than the actual sequence of occurrence of fraudulent and genuine transactions.
2. One might be interested in knowing how many matches India won in the last twenty test matches against Australia rather than a sequence of wins and defeats.
3. An E-commerce website may like to know the number of repeat orders rather than a sequence of orders.
4. A professor may like to know that how many classes were missed by a student rather than the sequence of presence and absence.
5. A social media influencer may like to know the total number of likes in his/her last 100 instagram reels rather than a sequence of likes and dislikes.

What is a special observation in the aforementioned scenarios? The answer is, in all these cases, we are not interested in the sequence of all the outcomes of a random experiment. Rather we are interested in knowing that how many times a particular outcome appears. The numerical measure of a particular outcome in a random experiment is known as a *random variable*.

Let $(\Omega, \mathcal{F})$ be a measurable space. A finite, single-valued function X that maps Ω into $\mathbb{R}$ (i.e., $X : \Omega \rightarrow \mathbb{R}$) is called a random variable if the inverse images under X of all Borel sets in $\mathbb{R}$ are events, i.e.,

$$X^{-1}(B) = \{\omega : X(\omega) \in B\} \in \mathcal{F} \quad \forall B \in \mathcal{B}. \quad (1)$$

To verify whether a real-valued function on $(\Omega, \mathcal{F})$ is a random variable, it is not necessary to check that (1) holds for all Borel sets $B \in \mathcal{B}$. It suffices to verify (1) for any class $\mathcal{C}$ of subsets of $\mathbb{R}$ that generates $\mathcal{B}$. By taking $\mathcal{C}$ to be the class of semiclosed intervals $(-\infty, x], x \in \mathbb{R}$, we get the following result:

Result: X is a random variable if and only if for each $x \in \mathbb{R}$,

$$\{\omega : X(\omega) \leq x\} = \{X \leq x\} \in \mathcal{F}. \quad (2)$$

Example 1: Let $\Omega = \{W, L\}$, and $\mathcal{F}$ be the class of all subsets of Ω . Here W and L denote *win* and *defeat* of a team in a match. Define X by

$$X(W) = 1, X(L) = 0.$$

Then

$$X^{-1}(-\infty, x] = \{\omega : X(\omega) \leq x\} = \begin{cases} \phi, & x < 0, \\ \{L\}, & 0 \leq x < 1, \\ \Omega, & x \geq 1. \end{cases}$$

Here $\mathcal{F} = \mathcal{P}(\Omega)$ (power set of the sample space), and $X^{-1}(-\infty, x] = \{\omega : X(\omega) \leq x\} \in \mathcal{F}$. Thus X is a random variable.

Example 2: Let $\Omega = \{WW, WL, LW, LL\}$, and $\mathcal{F}$ be the class of all subsets of Ω . Here W and L denote *win* and *defeat* of a team in two consecutive matches respectively. Define X by

$$X(\omega) = \text{Number of W's in } \omega.$$

Then $X(WW) = 2$, $X(WL) = X(LW) = 1$, $X(LL) = 0$.

$$X^{-1}(-\infty, x] = \{\omega : X(\omega) \leq x\} = \begin{cases} \phi, & x < 0, \\ \{LL\}, & 0 \leq x < 1, \\ \{LL, LW, WL\}, & 1 \leq x < 2, \\ \Omega, & x \geq 2. \end{cases}$$

Here $\mathcal{F} = \mathcal{P}(\Omega)$, and $X^{-1}(-\infty, x] = \{\omega : X(\omega) \leq x\} \in \mathcal{F}$. Thus X is a random variable.

Example 3: For any set $A \subseteq \Omega$, define

$$I_A(\omega) = \begin{cases} 0, & \omega \notin A, \\ 1, & \omega \in A. \end{cases}$$

$I_A(\omega)$ is called the indicator function of set A . I_A is a random variable if and only if $A \in \mathcal{F}$.

Remark 1: The notion of probability does not enter into the definition of a random variable.

Remark 2: Let $(\Omega, \mathcal{F})$ be a discrete sample space, i.e., Ω be a countable set of points and $\mathcal{F}$ be the class of all subsets of Ω . Then every numerical-valued function defined on $(\Omega, \mathcal{F})$ is a random variable.

Remark 3: Let X be a random variable defined on $(\Omega, \mathcal{F})$ and let a, b be constants. Then $aX + b$ is also a random variable on $(\Omega, \mathcal{F})$. Moreover, X^2 is a random variable and so also is $1/X$, provided that $\{X = 0\} = \phi$.